

HFT Liquidation Signal — Best Results (Shaw Team)

Trade filter for Binance maker fills. Metric: $\text{Score}(\tau) = \text{PnL_kept}(\tau) - \text{PnL_all}(\tau)$ (bps, higher is better). Out-of-sample on the FULL validation month (Feb 2026, every BTC trade).

Method

HistGradientBoosting markout regressor (one per horizon), trained on Dec 2025–Jan 2026, filtering trades with negative predicted markout (cutoff = 0, fixed in advance — not tuned on validation). Causal features (read $[t-W, t)$ only, Bybit liquidations shifted +200 ms): combined liquidation signed / absolute notional over 30·120·300 s, liquidation count, time-since-liquidation, BBO spread & book imbalance, 5 s mid-return, taker side, and the maker-adverse interaction $\text{side} \times \text{signed-flow}$. Markout labels reproduce the EDA baseline to ± 0.002 bps.

Results — BTC, validation = Feb 2026, ALL trades (~210 M), exact day-by-day accounting

τ	PnL_all (baseline)	PnL_kept	Score	filtered	kept turnover/day
30 s	−0.188	+0.193	+0.381	79%	\$3.1 B
120 s	−0.174	−0.292	−0.118	51%	\$7.2 B
300 s	−0.156	+0.073	+0.228	51%	\$7.1 B

Headline: the filter converts the toxic baseline into positive kept markout out-of-sample. Strongest at $\tau = 30$ s: **Score = +0.38 bps** (PnL_all −0.19 → PnL_kept +0.19). The Score-vs-aggressiveness curve is monotone for 30 s & 300 s; a more aggressive cutoff (+1.0 bps) lifts **Score(30 s) to +0.77** and **Score(300 s) to +0.34** while still keeping >\$1 B/day. The \$500k/day turnover constraint is never binding (kept turnover stays in the billions).

Links (GitHub)

- Task 3 — signal & full OOS evaluation: https://github.com/M-VS-V/hft-liquidations-2026/tree/main/task_3_signal_shaw_team
results log: `.../task_3_signal_shaw_team/results.txt`
- Task 2 — feature infrastructure (MarketContext, markout, causal windows):
https://github.com/M-VS-V/hft-liquidations-2026/tree/main/task_2_features_shaw_team
- Task 1 — EDA (baseline markouts, cascades): https://github.com/M-VS-V/hft-liquidations-2026/tree/main/task_1_eda_shaw_team